Sumit Kumar Kar

Data Scientist & Research Assistant

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Experience

•Research Assistant *March 2024 - Present*

DIAL: Design Inclusion and Access Lab

- Conduct comprehensive literature reviews to identify research gaps and determine suitable methodologies.
- Preprocessed and annotated image data to build research datasets.
- Leverage NLP analysis, along with qualitative data analysis and image processing, to derive valuable insights.
- Performed NLP-based sentiment and emotion analysis on interview transcripts and user-generated data.
- Develop data visualizations to effectively communicate findings using Python libraries.
- Contribute to writing research papers and preparing presentations for conferences and journal publications.
- Collaborate with multidisciplinary teams, including social science and computer science researchers, to integrate diverse perspectives into the research process.

•Research Assistant *February 2025 - Present*

North South University

- Co-lead a project, overseeing task delegation, team coordination, and milestone tracking.
- Conducting literature review to identify research gaps in AI-driven mental health support.
- Designed semi-structured interview questions for user-centered studies.
- Transcribed participant interviews conducted in both Bangla and English.
- Conducted preliminary thematic analysis to identify patterns.
- Develop an empathetic mental health chatbot powered by Large Language Models (LLMs).
- Collect and curate extensive datasets to fine-tune the chatbot for improved contextual understanding.
- Utilize fine-tuning and prompt engineering techniques to enhance chatbot responsiveness and empathy.
- Implement model optimization techniques to reduce computational requirements.
- Work under the supervision of faculty from both Computer Science and Psychology backgrounds, integrating interdisciplinary insights into model development.

•Research Team Lead *May 2025 – August 2025*

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- Leading research team in planning fieldwork, including field visit strategies and participant recruitment.
- Conducted 15 semi-structured interviews to uncover participant needs, practices, and systemic issues.
- Co-led 3 focus group discussions with stakeholders to gather diverse perspectives and validate insights.
- Collaborating with a multidisciplinary team to shape product ideas during the discovery phase.
- Transcribed and extracted key insights from qualitative interviews to inform design and strategy.
- Facilitating regular meetings with team members and the founder.
- Reporting progress and aligning findings with strategic direction through leadership communication.

Accepted Papers

- **Meena Needs a Computer Now: Co-design Solutions to Barriers using Fictional Inquiry for Women in Computing**
  ACM Transactions on Computer-Human Interaction (TOCHI) *August 2025*

- **Rokkha: Enabling Offline Emergency Location Sharing via SMS**
  ICCA: International Conference on Computing Advancements *June 2025*

- **A Novel Approach for Recognizing Real-Time American Sign Language (ASL) Using the Hand Landmark Distance and Machine Learning Algorithms**
  IEEE International WIE Conference on Electrical and Computer Engineering *November 2023*

Personal Projects

- **NARI: Women's Healthcare Smartphone Application for Bangladeshi Women**

  - Developed the app using Kotlin, ensuring concise syntax and enhanced safety for clean and efficient code.
  - Integrated Firebase Authentication for secure user login and registration processes.
  - Employed Firebase Firestore for flexible data management and real-time database functionalities.
  - Technology Used: Kotlin, Jetpack Compose, Material Design 3, Firebase, SQLite.

Education

- **Bachelor of Science in Computer Science and Engineering** *2020 - 2024*
  North South University, Dhaka CGPA: 3.17

Certifications

- **Introduction to Informed Consent**
  The Global Health Network February 2025
- **AWS Educate Machine Learning Foundations**
  Amazon Web Services June 2025
- **Introducing Generative AI with AWS**
  Udacity June 2025

Technical Skills and Domain Interests

Programming Languages: C/C++, Python, Java, Kotlin, JavaScript, HTML, CSS
Libraries & Frameworks: NumPy, Pandas, HuggingFace, Matplotlib, Seaborn
Tools & Software: Git, GitHub, LaTeX, Microsoft Office (Word, Excel, PowerPoint)
Database: Firebase, MySQL, SQLite
Relevant Courses: Machine Learning, Pattern Recognition, Neural Networks
Domain Interests: Artificial Intelligence, Natural Language Processing, Large Language Models

Research Focus Areas

Human-Computer Interaction (HCI) : Exploring user-centered design and technology adoption for diverse populations.
Natural Language Processing (NLP) : Applying natural language processing techniques for various AI applications.
Large Language Models (LLMs) : Fine-tuning and prompt engineering of LLMs for domain-specific solutions.
AI, ML & Deep Learning : Developing models for detection and prediction tasks across multiple domains.
Computer Vision & Image Processing : Developing methods to extract insights from visual and image-based data.
Mixed-Methods Data Analysis : Extracting actionable insights from qualitative and quantitative data
Qualitative Research : Conducting FGDs and interviews to understand user needs and inform design.
Interdisciplinary Research : Collaborating across domains to address complex, real-world problems.
AI and Policy Research : Exploring the intersection of artificial intelligence and policy-making
Social Science Research : Understanding human behavior, societal needs, and technology adoption.
Inclusive Technology Design : Collaborating with various underrepresented groups, including tertiary students, women in computing, people with disabilities, and micro-entrepreneurs to develop accessible, context-aware solutions that support wellbeing and inclusion.

Soft Skills

Soft Skills: Leadership, Teamwork, Problem-solving, Fast Learner, Communication, Adaptability, Time Management, Critical Thinking, Project Management, Research Skills, Presentation Skills, Collaboration

Language Proficiency

Bengali: Native language, fluent in speaking and writing.
English: Fluent in speaking, reading, and writing. Proficient in technical and academic writing.